ML-DL-Ops : Lab 2 Worksheet

Link of the colab notebook :
https://colab.research.google.com/drive/1MhTrK6o7Yx_oPJ4jNacIbOh-IQg3xnES?usp=sharing
Link for wandb :
https://wandb.ai/shikhar_dave-iit-jodhpur/cifar10-cnn-assignment

Introduction

This project implements a Convolutional Neural Network (CNN) for image classification on the CIFAR-10 dataset. The implementation includes a custom DataLoader, computational complexity analysis through FLOPs counting, comprehensive gradient flow visualization, and experiment tracking using Weights & Biases (WandB).

Dataset & Custom DataLoader

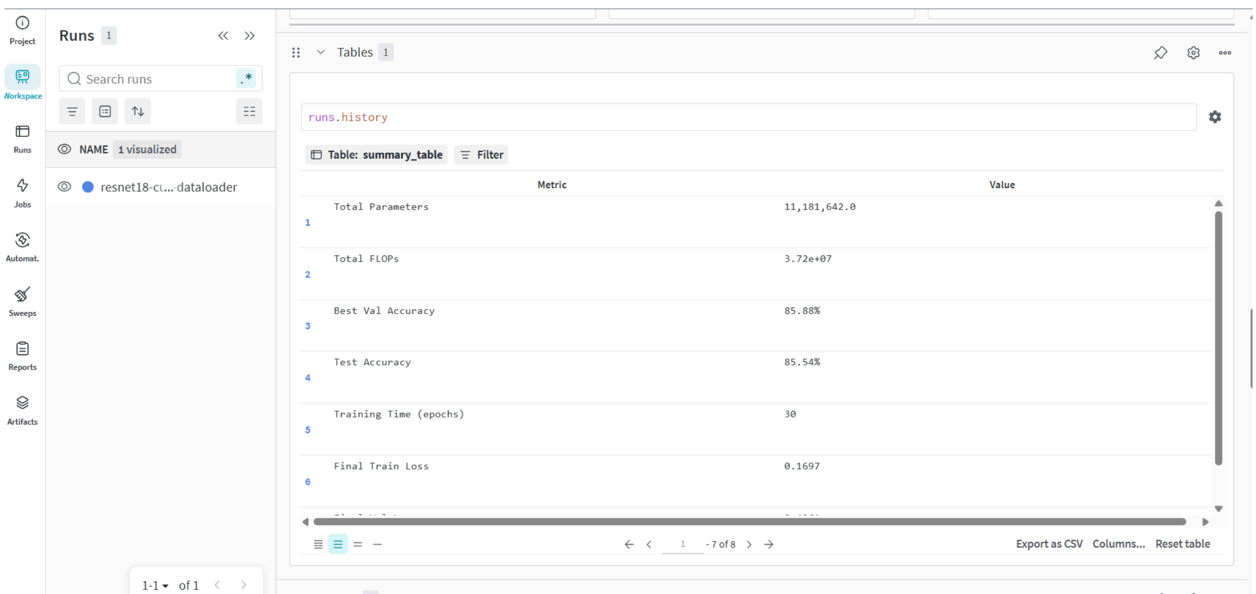
Dataset: CIFAR-10 (60,000 images: 50,000 training, 10,000 test)
Classes: 10 (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck)
Image Size: 32×32 pixels, RGB

Custom DataLoader Implementation: Developed a CustomCIFAR10Dataset class inheriting from torch.utils.data.Dataset with custom data augmentation pipeline including random cropping (padding=4) and horizontal flipping for training data. Normalization applied using CIFAR-10 statistics (mean=[0.4914, 0.4822, 0.4465], std=[0.2470, 0.2435, 0.2616]).

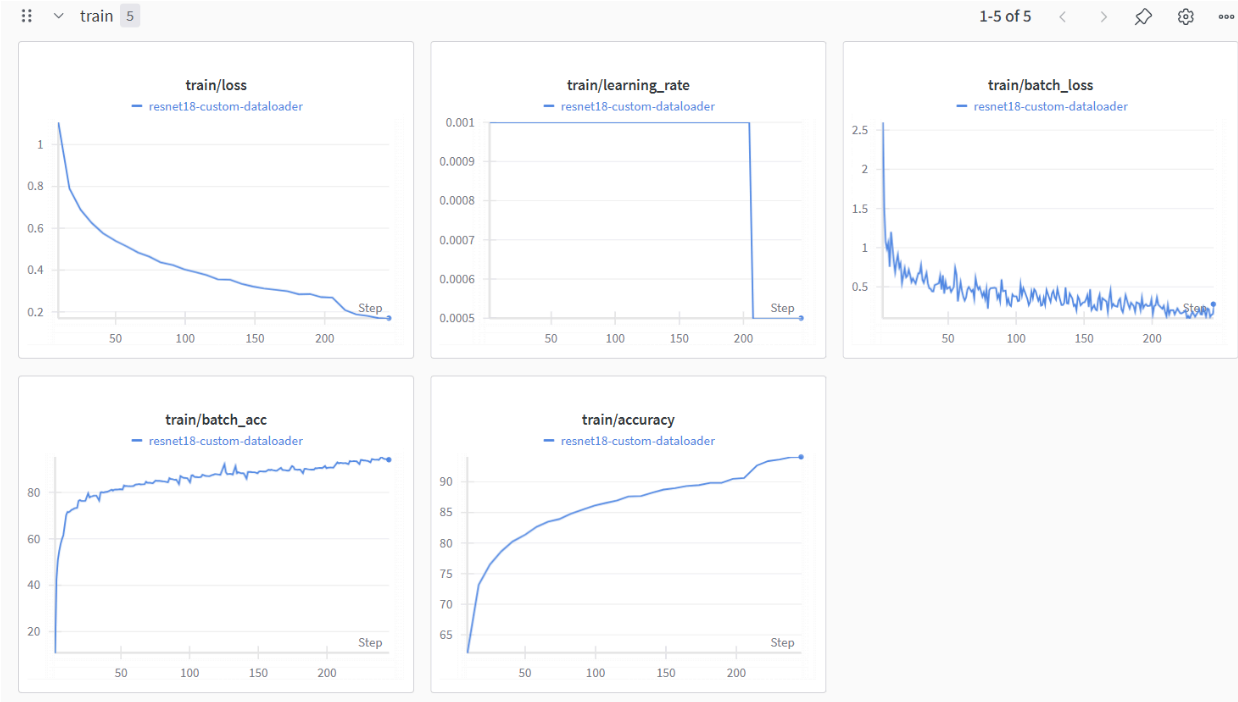
Model Architecture

Model: Pre-trained ResNet18 (ImageNet weights) fine-tuned for CIFAR-10
Modification: Final fully connected layer adapted for 10-class classification
Activation: ReLU
Architecture: 18 layers with residual connections (skip connections)

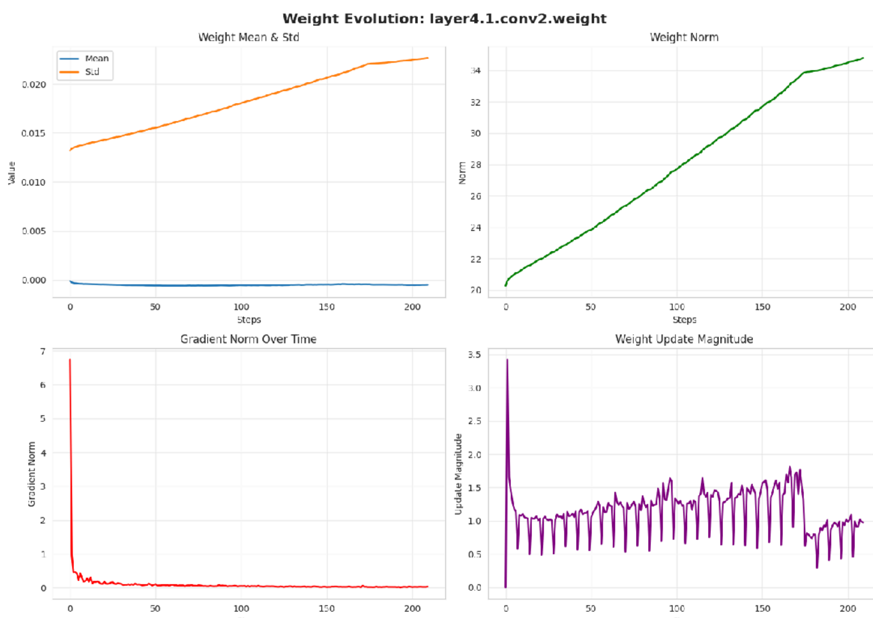
Important results and plots :

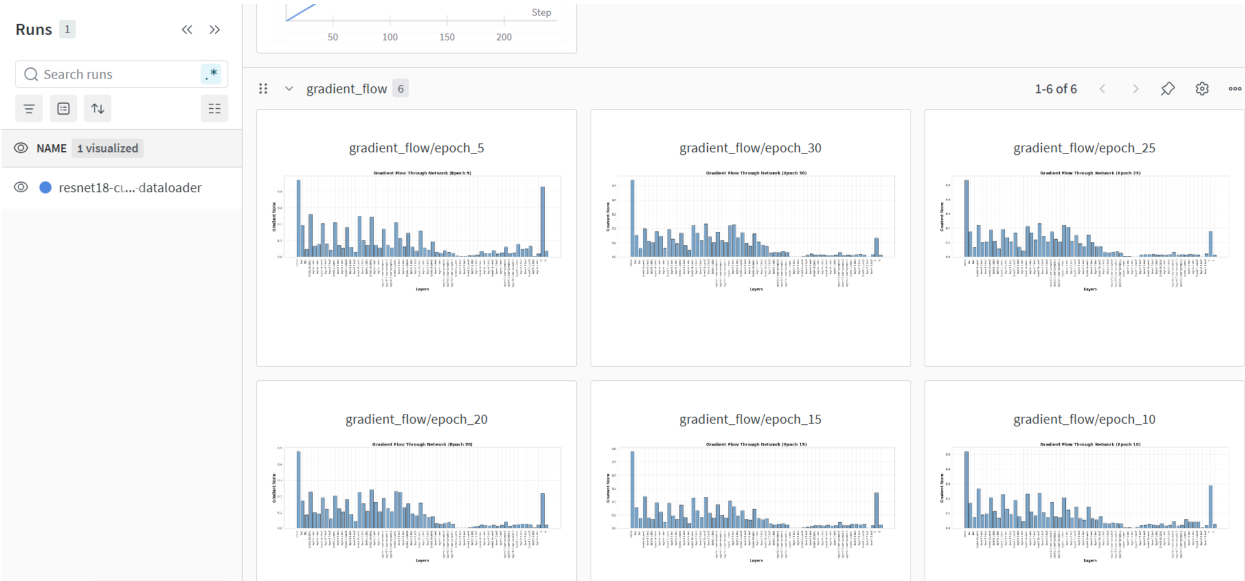


	Metric	Value
1	Total Parameters	11,181,642.0
2	Total FLOPs	3.72e+07
3	Best Val Accuracy	85.88%
4	Test Accuracy	85.54%
5	Training Time (epochs)	30
6	Final Train Loss	0.1697



weight_evolution/layer4.1.conv2.weight / resnet18-custom-dataloader




Runs 1
Search runs
.*
NAME 1 visualized
resnet18-cu... dataloader
Step
50
100
150
200
gradient_flow 6
1-6 of 6
gradient_flow/epoch_5
gradient_flow/epoch_30
gradient_flow/epoch_25
gradient_flow/epoch_20
gradient_flow/epoch_15
gradient_flow/epoch_10